

MSDM CEP Chapter 2 Development Progress Reports 1–5

Client: Restaurant at Kellogg Ranch

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1 Progress Report 1

1.1 Evidence Search and Approach Challenge

- [Google Drive Containing Individual Reports](#)

2 Progress Report 2

2.0.1 AO1. Newsletter Design and Reservation Growth (Leeann Lewis-Waddell)

Table 1: AO1 Model Comparison Table

Approach / Model	What It Explains	Evidence for / Against	Data / Assumptions	Fit with AO
Email Response Process (Hartemo, Suomi, and Hakala 2016; Kumar 2021)	How readers move from viewing a newsletter to clicking and eventually purchasing.	Hartemo, Suomi, and Hakala (2016) tested newsletter formats against a control, and Kumar (2021) found clicks had the strongest link to purchases. Both studies were conducted in commercial settings rather than a university newsletter.	Requires page views, clicks, and reservation data. Assumes each stage can be measured separately.	Strongest overall fit because it matches the stages Canva and OpenTable can measure.
Visual Attention and Heuristics (Kumar and Salo 2018; Skourou and Spiliotopoulos 2024)	Where readers look in a newsletter and which elements they are most likely to click.	Kumar and Salo (2018) found upper-left links earn higher click-through rates, and Skourou and Spiliotopoulos (2024) found large images capture attention. Images may also pull attention away from the button.	Requires versions that differ in button placement or media while all other elements stay the same.	Good fit for choosing the test variables, but does not connect clicks to reservations.
Cognitive Load and Information Overload (Bashirzadeh, Mai, and Faure 2022; Kong, Zhu, and Konstan 2021)	Why adding too much content, or sending too many emails, can reduce engagement.	Bashirzadeh, Mai, and Faure (2022) found combining visual elements increased unsubscribes, and Kong, Zhu, and Konstan (2021) found university recipients often ignore bulk email.	Requires page views alongside clicks to separate a design problem from a readership problem.	Useful boundary condition, but explains limits rather than what drives clicks.

Approach / Model	What It Explains	Evidence for / Against	Data / Assumptions	Fit with AO
Industry Best Practices (Go 2024; Jones 2023)	Practical recommendations for email design and benchmark click-through rates.	Both guides recommend strong visuals, bold CTAs near the top, and testing one variable at a time. Both are written by companies that sell marketing or reservation services.	Assumes general benchmarks apply to RKR's audience.	Helpful for benchmarks, but too weak to serve as the main framework.

The Email Response Process is the strongest fit for AO1 because it follows readers from viewing the newsletter to clicking the “Reserve a Table” button to making a reservation. This sequence matches the data available through Canva and OpenTable, which allows each stage to be measured. Visual attention research supports the decision to test button placement and media, while cognitive load research explains why only one change will be tested at a time. As such, the combination of these frameworks fits AO1 better than relying on industry best practices alone.

The evidence shows that button placement and visual design both affect click-through rate, and that clicks are closely connected to purchases. What remains uncertain is whether these findings will hold for a university audience that may already be overwhelmed by bulk email. It is also unclear whether the mailing list can be segmented and whether OpenTable can attribute reservations to separate links. Since two of my teammates are analyzing Facebook and Instagram, this analysis will focus only on the newsletter and should not assume it is the only channel influencing reservations. AO1 was already revised to test button placement in Fall and media in Spring, so no further revision is needed at this stage.

2.0.2 AO2. Customer Experience Factors Associated With Overall Satisfaction (Carmen Jimenez)

Table 2: AO2 Model Comparison Table

Approach / Model	What It Explains	Evidence for / Against	Data / Assumptions	Fit with AO
Multidimensional Restaurant Quality Framework (Ryu, Lee, and Kim 2012)	Explains how food quality, service quality, and the physical environment relate to customer satisfaction and behavioral intentions.	Ryu, Lee, and Kim (2012) support evaluating several restaurant-quality dimensions together. The original restaurant context differs from RKR.	Requires guest ratings of food quality, service quality, physical environment, and satisfaction. Assumes guests can evaluate these dimensions separately.	Strongest overall fit because it directly includes the three main experience dimensions measured in AO2.
Food Quality–Satisfaction Approach (Namkung and Jang 2007)	Explains how food quality relates to customer satisfaction and behavioral intentions.	Namkung and Jang (2007) support food quality as an important part of restaurant satisfaction. The approach focuses mainly on food rather than the entire dining experience.	Requires guest ratings of food-related characteristics such as quality, presentation, enjoyment, and consistency.	Good fit for the food-quality component of AO2, but too narrow to address the full objective alone.
Physical Environment / Atmospherics Approach (Han and Ryu 2009; Heung and Gu 2012)	Explains how atmosphere and the physical environment relate to satisfaction and behavioral responses.	Han and Ryu (2009) and Heung and Gu (2012) support the importance of the restaurant environment. This approach does not fully account for food or service quality.	Requires guest evaluations of atmosphere and physical-environment characteristics such as cleanliness, comfort, décor, layout, and overall ambiance.	Good fit for the atmosphere and physical-environment component, but not sufficient by itself to address AO2.

Approach / Model	What It Explains	Evidence for / Against	Data / Assumptions	Fit with AO
Customer Satisfaction–Behavioral Intentions Approach (Arora and Singer 2006)	Explains how customer satisfaction relates to future intentions such as returning to or recommending a restaurant.	Research supports a relationship between satisfaction and behavioral intentions, but stated intentions do not confirm actual future behavior.	Requires measures of overall satisfaction, intention to return, and willingness to recommend.	Useful secondary fit because AO2 includes return intention and recommendation, but it does not directly identify the strongest and weakest experience areas.

The Multidimensional Restaurant Quality Framework is the strongest fit for AO2 because it directly includes the three main experience dimensions measured in this objective: food quality, service quality, and the physical environment. Ryu, Lee, and Kim (2012) show that these dimensions are connected to customer satisfaction and behavioral intentions. This framework is broader than the food-quality and physical-environment approaches because it allows the dining experience to be evaluated as a combination of multiple factors rather than focusing on only one area.

The existing research shows that food quality, service quality, and atmosphere or physical environment are important parts of the restaurant experience. However, the literature cannot determine which of these areas RKR guests rate most positively or negatively. This is especially important because RKR is a student-operated restaurant located on a university campus, which may create a different guest experience from the restaurants studied in previous research. For this reason, AO2 still requires primary survey data from RKR guests. The current AO does not need to be revised because it is measurable, directly connected to the available survey data, and aligned with the project objective of identifying strengths and improvement opportunities in the RKR dining experience.

2.0.3 AO3 Facebook Content and Messaging Strategies and Reservation Performance

Table 3: AO3 Model Comparison Table

Approach/Model	What It Explains	Evidence for/Against	Data/Assump- tions	Fit with AO
Processing Fluency (Philp, Jacobson, and Pancer 2022a)	Why clear and recognizable visual content may be easier for customers to process and respond to	Recognizable food images received stronger engagement but does not explain reservation	Posts have to be grouped by visual/content characteristics and compared using engagement metrics.	Good fit for explaining visual content performance but not enough for the full AO
Media Richness+Speech Act Theory (Manningham, Asselin, and Bourguignon 2024)	How wording, format, and message presentation might affect engagement	Direct wording and emoji use were linked to higher restaurant Facebook engagement	Posts would need to be coded by message style, wording, and format.	Good fit for messaging section of AO but not the reservation part
Digital Consumer Engagement+Customer Behavioral Response [Halloran and Lutz (2021); (Yost, Zhang, and Qi 2021)]	How customers engage with Facebook content and how that engagement can be related to customer behavior	Stronger engagement was linked to purchase frequency and sales.	Would need Facebook engagement and a downstream behavioral measurement like OpenTable activity	Overall this would be the best fit because AO3 is examining Facebook engagement and reservation-related behavior
Theory of Reasoned Act/Customer Journey [Hanaysha (2022); (Velooso and Gomez-Suarez 2023)]	How customers might move from engagement stage to the action stage	Engagement was linked to purchase intention but intention is not an actual measurable behavior	This assumes that outside factors might exist between the initial engagement stage to the reservation stage.	Weaker than the others for analyzing reservation activity

Digital Consumer Engagement and Customer Behavioral Response are the strongest fits for AO3 because

they are directly connected to Facebook engagement with downstream behavior. Digital Consumer Engagement helps explain how customers react to different Facebook content and messaging, while Behavioral Response helps connect the initial engagement to an action that is taken later like making a reservation or visiting the RKR. This combination fits AO3 better than the others because they are more useful for explaining why specific posts perform better, and intention, but don't explain the following steps in the customer journey.

The evidence show that Facebook content, visual presentation, and messaging style can influence customer engagement. It also shows that some forms of engagement might be connected to customer behavior. What is uncertain is whether Facebook strategies that perform better for RKR will be connected to stronger OpenTable reservation activity. One thing to also keep in mind is that two of my group members are analyzing Instagram and the newsletter, so this analysis will only focus on Facebook and should not assume that Facebook is the only channel influencing reservation. At this point, AO3 does not need to be revise because it fits the overall project and available research.

2.0.4 AO4 Website Content Navigation and Usability (Georgia Monjarrez)

Approach / Model	What It Explains	Evidence for / Against	Data / Assumptions	Fit with AO
Selected DeLone and McLean constructs (DeLone and McLean 2003)	Separates information/system quality visitor responses and benefits.	Broad established framework. does not prove effects of RKR changes.	Content and navigation review feedback defined use outcomes. full model testing requires adequate measures and sample.	Organizing framework with task performance as the project outcome.

Approach / Model	What It Explains	Evidence for / Against	Data / Assumptions	Fit with AO
Technology acceptance and hospitality information quality (Davis 1989; Jeong and Lambert 2001)	Distinguishes perceived ease usefulness and information quality.	Supports separate visitor perceptions. original technology and simulated hotel contexts limit transfer.	Adapted five-point ease item and optional usefulness feedback. adapted items are not the complete validated scales.	Useful supplement. perceptions alone cannot establish the 80% completion target.
DINEWE-BQUAL and information-task fit (Dedeke 2016; Kim et al. 2012)	Restaurant-specific quality and information relevant to tasks.	Restaurant dimensions supported in 211 students. travel context supports task fit. Intentions do not establish bookings.	Review categories and feedback on menu/hours/reservation tasks. adaptations require caution.	Guides content and quality diagnosis alongside task observation.
Task-based usability evaluation (Hornbæk 2006; Hornbæk and Law 2007; Tuch et al. 2012)	Whether visitors complete tasks time needed and experienced difficulty.	Supports distinct performance and perception measures. small samples limit generalization and appearance is insufficient.	Completion counts/rates median time for successful unassisted attempts and five-point ease distributions. defined endpoints and recorded devices/familiarity.	Principal method for evaluating each task against the proposed 80% target. selected accessibility checks supplement it.

AO4 Model Comparison Table

2.0.5 AO5 Instagram Content Format and Engagement (Spencer Conway)

Approach / Model	What It Explains	Evidence for / Against	Data / Assumptions	Fit with AO
Stimulus–Organism–Response (S–O–R) Framework (Bushara et al. 2023)	How social media content can affect what people think and how they respond.	Restaurant research supports connections between social media marketing, perceived value, and behavioral intentions. It does not show which Instagram format gets the most engagement.	Would need audience feedback to measure perceptions, along with engagement data. Assumes those perceptions help explain the response.	Good support for explaining audience responses, but Instagram Insights cannot test the whole framework.
Processing Fluency [(Philp, Jacobson, and Pancer 2022b)]	Why clear and recognizable food images may be easier to understand and engage with.	Typical-looking food images were linked to stronger engagement, but this does not tell us whether Reels, carousels, or single images work best.	Posts would need to be grouped by visual characteristics and compared using engagement metrics. Audience feedback would be needed to test the explanation directly.	Helpful for the visual content part of the AO, but not enough to answer the full objective.
Uses and Gratifications Theory (Kulikovskaja et al. 2023)	Why people engage with content that provides information, entertainment, or social connection.	Restaurant research connects different content types with engagement. However, findings from other platforms may not apply directly to RKR’s Instagram.	Posts would need to be grouped by content purpose as well as format. Assumes people choose content that meets their needs; audience feedback would help confirm those motivations.	Useful for explaining why people respond to content, but does not identify the strongest format by itself.

Approach / Model	What It Explains	Evidence for / Against	Data / Assumptions	Fit with AO
Post-Format Comparison and Content Testing	Which Instagram formats get stronger engagement for RKR.	Directly addresses the AO, but differences could also come from the topic, posting time, or paid promotion.	Requires post format, reach, likes, comments, shares, and saves. Posts need similar measurement windows and reasonably similar posting conditions.	Best overall fit because AO5 looks at past posts and then tests new content in selected formats.

AO5 Model Comparison Table

Post-format comparison with support from S–O–R and Uses and Gratifications is the best fit for AO5. Comparing formats will help us see whether Reels, carousels, or single-image posts get stronger engagement for RKR. S–O–R helps explain how content might affect audience responses. Uses and Gratifications adds another explanation by looking at what people get from the content, such as information, entertainment, or social connection. Processing fluency helps explain why some food images may perform better. These theories do not tell us which format is best for RKR, so we still need to look at the account’s actual results. Past posts will be grouped by format to create a baseline, and new posts will be compared with those results. Engagement rate will be calculated using likes, comments, shares, and saves divided by reach, multiplied by 100.

The research suggests that both visual presentation and content purpose can affect engagement. Kulikovskaja et al. (2023) found different engagement patterns for informational and entertaining, reward-based, and relationship-focused restaurant content. However, the research does not establish what works best for RKR’s Instagram audience. We will need Instagram Insights data and information about each post’s topic, purpose, posting date, and whether it was promoted. For example, a Reel might perform bet-

ter because it is entertaining, rather than just because it is a video. New posts should be measured over the same amount of time so the comparison is more useful. Stronger engagement also does not automatically mean more reservations. At this point, AO5 does not need a major revision, although adding “among the formats tested” would make it more specific. We still need faculty guidance on how far back to look at past posts and how many new posts to test.

2.1 Investigation of Currently Available Research Framework/Model

2.1.1 AO1. Newsletter Design and Reservation Growth (Leeann Lewis-Waddell)

Analytics Objective: Compare “Reserve a Table” click rates and attributed OpenTable reservations across RKR newsletter section designs, testing button placement in Fall 2026 and image versus video in Spring 2027.

Existence of Well-established Models

Analytics Objective 1 intends to A/B test revisions of the current advertising for the Restaurant at Kellogg Ranch within The Collins Concierge, the department’s weekly newsletter. Specifically, it will test if an image-based or video-based ad will increase “Reserve a Table” clicks in comparison to the current text-only format. After extensive research, three established frameworks inform how that test will be built and executed. AO1 will examine the email response process, visual attention and heuristics, and cognitive load and information overload.

Email marketing response occurs in a sequence, wherein a reader first sees a message, engages with it, and then chooses whether to act on it. Hartemo’s newsletter testing model focuses on this response process, linking the factors that affect response to the metrics that measure each step (Hartemo, Suomi, and Hakala

2016). This study tests eight format variations against a control group. The format of this analysis easily applies directly to the comparison of image and video templates against RKR's current section, which employs a controlled format test with a no-graphic baseline.

A review of Kumar's framework divides newsletter success into relational metrics, such as opens and clicks, and transactional metrics, such as purchases (Kumar 2021). Drawing on 811 newsletters sent to 19,678 loyalty program members, the study determines that design significantly shaped response, and clicks had the strongest relationship to eventual purchase (Kumar 2021). This point supports using click-through rate as AO1's primary metric, with reservations as the secondary, validating measure. With respect to RKR, Kumar's sequence would apply to three measurable stages: newsletter views in Canva, "Reserve a Table" clicks, and OpenTable reservations.

One point of note: the response process also exposes a gap AO1 has to account for. Since RKR welcomes walk-ins and accepts Bronco Bucks, some readers will move from acknowledging the new template section and move directly to a visit without ever clicking. Therefore, version-specific links or perhaps a "How did you hear about us?" question at booking help capture that path, but would require additional points in the customer journey to address this variable.

As can be expected, readers do not scan a newsletter evenly or in its entirety. Described in detail by Kumar, visual heuristics guide where the eye lands and subsequently what gets clicked. This study details how newsletter readers follow a U-shaped pattern rather than the Z-pattern typical of web pages. It also outlines how the upper-left region produces higher CTR than links placed elsewhere (Kumar and Salo 2018). In the current RKR section, "view the menu" is a clickable link at the top center while the "Reserve a Table" button sits at the bottom, below the hours. Because placement carries its own documented effect, both test versions keep the button in the same position; otherwise a change in clicks could come from placement rather than media. Using current metrics as a baseline, AO1 will test relocation of the clickable

“Reserve a Table” button for the remainder of the fall, and media additions through the sixteen weeks of Spring.

With regard to adding media to the RKR section of The Collins Concierge, Imagery is known to compete strongly for attention once an email is open. Skourou’s study using eye-tracking technique and tools found that large images captured more attention than discount codes and prices in promotional emails. Notably, what readers chose to look at mapped to their purchase decisions (Skourou and Spiliotopoulos 2024). This supports visual content as AO1’s main test variable, since the current section depends on a block of text to communicate its key details, specifically food and its accompanying atmosphere. On the contrary, the study also shows that a strong image or video could consequently draw focus away from the button it is meant to support.

Industry guides from Semrush and OpenTable support the aforementioned notion. High-quality images are most recommended, coupled with contrasting CTAs, and testing one variable at a time over at least two weeks (Go 2024). OpenTable’s restaurant guide adds that key information belongs at the top, supported by strong food photos and bold CTA buttons (Jones 2023). For RKR, Chef Matt’s seasonal dishes offer natural subject matter for either format.

This objective intends to be mindful of the potential for cognitive overload and the possibility to annoy or turn-off the target audience. Visual content can compete for a reader’s limited attention rather than enhance or entice. A large field experiment found that while GIFs and emojis did not affect CTR individually, combining them inadvertently increased unsubscribes, as the elements competed for the same cognitive resources (Bashirzadeh, Mai, and Faure 2022). As such, AO1 aims to add only one visual element at a time. The image and video templates will match in layout, copy, and button placement, making media the only variable.

Research for this objective also flags the potential for inbox overload. Several studies found university

recipients reported receiving 30 bulk emails in a single week, often opening them without reading and trashing 22% of newsletters (Kong, Zhu, and Konstan 2021). At one business school, academics received 324 organizational bulk emails in a single year with no option to unsubscribe (Bicudo De Castro 2025). As such, if Collins Concierge readers are facing similar volume in addition to academic workload, this objective must acknowledge that a redesign may not move CTR regardless of format. This is why AO1 measures clicks per view directly rather than relying on open rates.

Email also struggles when it lacks a clear role in the audience's decision process. Survey respondents reacted with resistance and irritation toward email marketing that did not fit their customer journey (Zunac, Miletic, and Kozina 2022). The results of this study justify treating the redesign as one input toward reservations rather than its sole driver.

Adequateness of Gathered Information

The information above is significantly adequate in addressing the low click-rate of RKR's newsletter channel. It addresses the potential roadblocks in attracting the attention need to gain awareness, which is overall design and aesthetics, particularly as it relates to the placement of the desired goal, which is button location, but also fatigue since the section does not change and does not employ eye-catching media. On the other hand, it raises red flags with regard to extenuating circumstances which may supercede the design optimization, which are out of RKR's control such as inbox overload. The studies align with industry suggestions, all of which are actionable and applicable to the business problem. The only biases are really in the industry guides from Semrush and OpenTable, since they are businesses providing free resources to sway readers into purchasing services (Go 2024), (Jones 2023). Additionally, the eye-tracking study had a very small pool of participants, and the Zunac study of the effectiveness of email as a single marketing channel was largely comprised of female participants, making the data somewhat biased (Skourou and

Spiliotopoulos 2024), (Zunac, Miletic, and Kozina 2022). This research will be applied to revising the RKR section of The Collins Concierge, by addressing button placement and aesthetics. Beginning with a baseline of the previous issue's metrics, Fall will test only relocation of the button to the upper left, and Spring will A/B test video versus image.

Need for Additional Analysis

Since the gathered information was largely sufficient but missing data specific to the Concierge's audience, additional analysis is needed to fully address the AO. None of the reviewed studies looked at a restaurant section inside a university newsletter, so primary data will fill that gap. Beginning with a baseline of CTR from previous Concierge issues pulled from Canva, Fall will measure the effect of moving the "Reserve a Table" button to the upper left of the current text-only section, and Spring will A/B test image versus video, alternating weekly with the button in the same position. Page views will be tracked alongside clicks, since the university studies suggest inbox overload could keep readers from ever reaching the RKR section; this will help determine whether low CTR is a design problem or a readership problem (Kong, Zhu, and Konstan 2021), (Bicudo De Castro 2025). Each version will also use its own link to OpenTable so reservations can be tied back to the format that drove them. On the other hand, clicks will not capture everyone, particularly guests who walk in or pay with Bronco Bucks, so a "How did you hear about us?" question at booking will help account for those visits. Holidays, finals, and campus events will be noted throughout, since alternating weekly means outside factors could affect results. Click rates between versions will be compared using a two-proportion test to confirm whether differences are meaningful.

Assessment and Affirmation of Analytics Objectives

The original AO set out to A/B test two RKR section formats in The Collins Concierge from October 2026 to June 2027, comparing "Reserve a Table" click rates and attributed OpenTable reservations between

versions. It still holds, but the research has narrowed it, particularly as it relates to what is being tested and when. It addresses RKR's need to build sustainable demand by improving a section that is currently easy to scroll past, with no imagery and a button placed at the bottom, below the hours. On the other hand, testing media and button placement at the same time would make it impossible to know which change caused an increase, so the AO will be split into two phases. Two other approaches were considered. A factorial test would compare every combination of button placement and media at once [Hartemo 2016]. However, it would require splitting the mailing list into four groups, leaving too few issues per version within one semester. A bundled redesign would be simpler, but it would not show which change made the difference. As such, the two-phase approach is the most practical option for isolating one change at a time. Beginning with a baseline of CTR from previous issues, Fall will test only the relocation of the button to the upper left, and Spring will A/B test video versus image, alternating weekly with the button in the same position. Canva will be used to track page views and clicks, which is necessary for separating a design problem from a readership problem, and each version will use its own link to OpenTable to tie reservations back to the format. A two-proportion test will compare click rates between versions, since it is a simple and appropriate way to determine whether a difference is real or due to chance. These tools are appropriate because they are already available to RKR and the Concierge team, they require no individual reader data, and they follow the same steps a reader takes from viewing the newsletter to clicking to booking (Hartemo, Suomi, and Hakala 2016).

2.1.2 AO2. Customer Experience Factors Associated With Overall Satisfaction (Carmen Jimenez)

Analytics Objective: Using anonymous post-visit survey responses, compare RKR guests' ratings of food quality, service quality, and atmosphere/physical environment to identify the strongest and weakest

experience areas.

Existence of Well-established Models

Customer satisfaction is generally understood as a guest's overall evaluation of a dining experience. In restaurant settings, this evaluation is usually influenced by several parts of the experience rather than by one factor alone. For AO2, the most relevant dimensions are food quality, service quality, and atmosphere or the physical environment. These dimensions provide a useful framework for understanding what RKR guests value most and where improvements may be needed.

A strong framework for AO2 comes from Ryu, Lee, and Kim, who examined how food quality, service quality, and the physical environment contribute to restaurant image, perceived value, customer satisfaction, and behavioral intentions (Ryu, Lee, and Kim 2012). This framework is especially useful for AO2 because it brings together the three main experience dimensions that RKR plans to measure. Instead of evaluating the dining experience only through food or only through service, the framework treats the restaurant experience as multidimensional. This supports the structure of the RKR survey and allows the team to compare different parts of the guest experience using the same rating scale.

For RKR, this multidimensional approach is important because the restaurant experience includes more than the meal itself. Guests interact with the food, the service staff, and the restaurant environment during the same visit. A guest may rate one of these areas very positively while rating another area less favorably. Looking at the dimensions separately can therefore provide more useful information than relying only on one overall satisfaction question. The framework also supports the use of overall satisfaction as a broader measure that can be considered alongside the individual experience dimensions.

Food quality is one of the most established components of restaurant satisfaction. Namkung and Jang found that food quality is associated with customer satisfaction and behavioral intentions in restaurant settings

(Namkung and Jang 2007). Food quality can include characteristics such as taste, freshness, presentation, temperature, portion size, and consistency. These characteristics help explain why food should be evaluated as its own part of the dining experience rather than being combined with service or atmosphere.

This approach is relevant to AO2 because the survey can ask RKR guests to evaluate their perception of the food experience directly. Ratings of food quality can then be compared with ratings of service quality and atmosphere or physical environment. If food receives the highest average ratings, it may represent a current strength of the restaurant. If it receives lower ratings than the other experience areas, the results may identify an opportunity for improvement. However, the food-quality approach alone is too narrow to fully address AO2 because the objective examines the overall dining experience through several dimensions.

Service quality is another important component of restaurant customer satisfaction. Ryu, Lee, and Kim's framework treats service quality separately from food quality and the physical environment (Ryu, Lee, and Kim 2012). This distinction is useful because a guest can enjoy the food while still having a less positive experience with service, or the opposite. Service quality may involve friendliness, attentiveness, responsiveness, professionalism, communication, and reliability.

Service quality is particularly relevant to RKR because the restaurant operates within The Collins College of Hospitality Management and also functions as a learning environment. Guests are still customers of the restaurant and evaluate the service they receive as part of their dining experience. Measuring service quality separately can help RKR identify whether guests view service as one of the strongest parts of the experience or whether certain aspects may need attention. For AO2, service ratings will therefore be compared with food-quality and atmosphere ratings rather than being interpreted in isolation.

Atmosphere and the physical environment represent another established area of restaurant research. Han and Ryu examined relationships among the physical environment, price perception, customer satisfaction, and restaurant loyalty (Han and Ryu 2009). Heung and Gu also examined the influence of restaurant

atmospherics on customer evaluations and behavioral responses (Heung and Gu 2012). Together, these studies support including the physical environment as a separate dimension of the dining experience.

The physical environment may include elements such as cleanliness, décor, lighting, seating, layout, comfort, and the overall atmosphere created by the restaurant. For RKR, this dimension may also reflect the restaurant's campus setting and the overall feeling guests experience during lunch or dinner service. These elements are different from the food and service themselves, but they can still contribute to how guests evaluate the visit.

The physical-environment approach is useful for AO2 because it allows RKR to identify whether the atmosphere is evaluated as positively as the food and service. However, similar to the food-quality approach, it is not sufficient by itself to answer the full analytics objective. AO2 requires a broader comparison of several experience dimensions. For this reason, the multidimensional restaurant-quality framework provides a stronger overall foundation, while the food-quality and physical-environment approaches provide additional support for individual dimensions.

Previous restaurant research also connects customer satisfaction with future behavioral intentions, including intention to return and willingness to recommend a restaurant (Arora and Singer 2006). These measures are relevant to the RKR project because positive dining experiences may contribute to repeat visits and positive recommendations. However, AO2 will treat intention to return and willingness to recommend as secondary descriptive indicators rather than as evidence of actual future behavior. A guest stating that they intend to return does not confirm that a repeat visit will actually occur.

Perceived value also appears in several restaurant models, including the framework used by Ryu, Lee, and Kim (Ryu, Lee, and Kim 2012). However, perceived value will not be treated as a primary dimension in the current AO2. The revised objective is intentionally focused on food quality, service quality, and atmosphere or physical environment. Keeping these three dimensions as the main areas of comparison

makes the objective more focused and directly connected to the planned survey.

Taken together, the existing research provides a strong conceptual foundation for AO2. The multidimensional restaurant-quality framework is the strongest overall fit because it directly includes the three experience dimensions that RKR plans to compare. The food-quality and physical-environment approaches provide additional support for understanding individual parts of the guest experience. These models will guide the design of the anonymous post-visit survey and the interpretation of the results.

The survey will use five-point rating items to measure food quality, service quality, atmosphere and physical environment, overall satisfaction, intention to return, and willingness to recommend. Open-ended questions will also allow guests to identify the best part of their experience and areas that could be improved. The existing research therefore helps determine what should be measured, but it cannot determine which experience areas are actually strongest or weakest at RKR.

This limitation is important because the previous studies were conducted in restaurant settings that differ from RKR. RKR is a student-operated restaurant located on a university campus and serves its own mix of guests. The existing models can guide the research design, but primary survey data from RKR guests are still necessary to understand the restaurant's specific customer experience. For that reason, the literature provides the framework for AO2, while the RKR survey will provide the evidence needed to identify strengths and improvement opportunities.

Adequateness of Gathered Information

The gathered information is sufficient to identify the main dimensions that should be measured for AO2, including food quality, service quality, atmosphere or physical environment, overall satisfaction, and future behavioral intentions. The reviewed studies are generally consistent in showing that these dimensions are important parts of the restaurant experience, and several of the sources are directly based on restaurant

settings (Han and Ryu 2009; Heung and Gu 2012; Namkung and Jang 2007; Ryu, Lee, and Kim 2012). However, the existing research is not sufficient to determine which dimensions RKR guests currently evaluate most positively or which areas need the most improvement. The studies were conducted in different restaurant contexts and populations, so their findings cannot be assumed to apply directly to RKR. Primary survey data from RKR guests are therefore necessary to complete the objective. Possible limitations of the planned survey include convenience sampling, nonresponse bias, and the possibility that guests with a connection to Cal Poly Pomona or RKR may provide more favorable responses. The existing literature will therefore be used to guide what is measured, while the RKR survey will provide the restaurant-specific evidence needed to identify strengths and improvement opportunities.

Need for Additional Analysis

Additional analysis is needed because the existing research cannot determine which parts of the RKR dining experience are currently strongest or weakest for its own guests. To address this gap, anonymous post-visit survey responses will be analyzed using response counts, percentages, and average ratings for food quality, service quality, atmosphere or physical environment, and overall satisfaction. Open-ended comments will also be reviewed and grouped into recurring themes to provide additional context about what guests value and what they believe could be improved. Intention to return and willingness to recommend will be summarized as secondary descriptive indicators. If the number of responses is sufficient, simple exploratory correlations may also be used to describe how the experience dimensions are associated with overall satisfaction. These analyses will not be interpreted as causal evidence. The additional primary-data analysis will allow RKR to apply the existing research framework to its own guests and identify specific customer-experience priorities that cannot be determined from the literature alone.

Assessment and Affirmation of Analytics Objectives

The current AO2 remains appropriate because it directly addresses RKR's need to understand which parts of the guest experience are performing well and which areas may require improvement. The objective is measurable using anonymous post-visit survey responses and is closely aligned with PO2, which focuses on identifying what RKR guests value most and where improvements are needed in the overall dining experience.

To accomplish AO2, the analysis will use descriptive statistics to compare guest ratings of food quality, service quality, atmosphere or physical environment, and overall satisfaction. Response counts, percentages, and average ratings will be used to identify the highest- and lowest-rated experience areas. Open-ended comments will also be reviewed and organized into recurring themes to provide context for the numerical results and identify specific strengths or improvement opportunities mentioned by guests.

Intention to return and willingness to recommend will be included as secondary descriptive indicators. If the sample size is sufficient, simple exploratory correlations may also be used to examine whether higher ratings in food quality, service quality, or atmosphere are associated with higher overall satisfaction. These relationships will be interpreted as associations rather than causal effects.

These analytical techniques are appropriate because they match the type of data expected from the survey and directly support the purpose of AO2. The descriptive analysis will allow RKR to compare major experience dimensions in a clear and actionable way, while the open-ended responses can help explain why guests rated certain areas more positively or negatively. The optional exploratory correlations can provide additional insight into which experience dimensions appear most closely associated with overall satisfaction. Together, these methods will help RKR identify customer-experience priorities and make evidence-based decisions about where improvements may have the greatest value.

2.1.3 AO3.Facebook Content and Messaging Strategies and Reservation Performance (Luis Ahumada)

Analytics Objective: Analyze associations between facebook content and messaging strategies, engagement metrics, and OpenTable reservation activity.

Existence of Well-established Models (2-3 pages per AO)

There are several types of models and theories that can support AO3 and help explain why different Facebook content and messaging may lead to different levels of customer engagement. The key ideas that are connected to AO3 are **digital consumer engagement**, **Media Richness**, **Speech Act Theory**, **processing fluency**, the **Theory of Reasoned Action**, and **customer behavioral response**. Together these ideas help explain how customers process social media content, how the way a message is presented can affect engagement, and why engagement does not always lead directly to an actual reservation. These theories are important to AO3 because the objective looks past simply identifying which Facebook posts receive the most engagement and also considers whether those patterns are connected to OpenTable reservation activity.

Processing fluency helps explain why visual representation of content may affect engagement. Philp, Jacobson, and Pance found that consumers responded better to food images that were easier to recognize and understand (Philp, Jacobson, and Pancer 2022a). Drossos, Coursaris, and Kagiouli's study found that Facebook posts using images and emotional or transformational appeals had a higher rate of engagement (Drossos, Coursaris, and Kagiouli 2024). Their study also showed that posts with higher interaction did not always perform better. Together, these findings support AO3 because they show that Facebook performance can depend on different visuals and content characteristics, not just the topic of the content or messaging.

Media Richness Theory and Speech Act Theory help explain why the way a message is communicated can play a factor in a customer's response. Manningham, Asselin, and Bourguignon performed a study on 639 Facebook posts from different restaurants and analyzed the wording, post format, and use of emojis (Manningham, Asselin, and Bourguignon 2024a). The finding was that direct and expressive wording, along with emoji use, was tied to higher customer engagement. Velosa and Gomez-Suarez also found that when management responses matched the original social media content, customers showed higher engagement and stronger booking intention (Veloso and Gomez-Suarez 2023). These findings support AO3 because they show that customer response can really depend on how content is written and presented, not just the type of content being posted. This means that when analyzing RKR Facebook content, it would make sense to compare both the type of content and the way the message within the content is communicated.

Digital consumer engagement is also important because it shows us that not every type of engagement has the same definition behind it. Halloran and Lutz found that stronger forms of engagement, like customers commenting on a post, were more closely related to restaurant purchase frequency. Weaker measures like likes had less of a relationship with restaurant visit frequency (Halloran and Lutz 2021a). This supports AO3 because it allows us to see that not every Facebook metric can be treated equally when deciding which strategies perform best. Yost, Zhang, and Qi also found that social media engagement was related to sales, but engagement did not explain all of the challenges in business performance (Yost, Zhang, and Qi 2021). These findings show that engagement can be useful for AO3, but it should not be used on its own to determine whether a specific Facebook marketing strategy is successful or not, instead tying it to reservation activity can help support the final decision.

The Theory of Reasoned Act and customer behavioral response explain why customer intention shouldn't be treated the same as actual behavior. Hanaysha's study found that interactivity, relevance,

informativeness, and entertainment were positively related to brand engagement, and the higher engagement was tied to higher purchase intention (Hanaysha 2022). So this supports the idea that social media content can affect consumer response, but the challenge is that the study measured intention instead of actual customer behavior. Simonetti and Bigne also found a similar result when they performed a study measuring restaurant-related social media cues. Their findings showed that restaurant images were important for capturing initial attention, while reviews and ratings also played a factor on how customers evaluated the restaurant (Simonetti and Bigne 2022). These findings challenge AO3 because they tell us that a customer can respond in a positive manner to a content being posted, but this doesn't necessarily mean that they intend to visit the restaurant without actually making a reservation. Again, this is why looking at more than just Facebook engagement is important for AO3, as it will strengthen the final decision on what Facebook Strategy works best.

A separate challenge that was brought up was that customer engagement might also be influenced by activity on other social media platforms. Unnava and Aravindakshan studied Facebook, Twitter, and Instagram and found that engagement on one platform could be affected by activity on another platform (Unnava and Aravindakshan 2021). Another finding was that the effects that each platform has on each other isn't limited to a specific amount of sessions, this effect can continue over time. This is connected to AO3 because it shows us that changes in Facebook engagement or reservation activity may not be connected to Facebook on its own. RKR customers might also be exposed to content on Instagram, Youtube, CPP Newsletters, CPP Website, etc. playing a role on the customers decision of visiting the restaurant or making the reservation. Because of this, Facebook metrics should be interpreted carefully when comparing with OpenTable activity.

Overall, the research supports the main idea behind AO3, but also shows that Facebook performance should not be judged by engagement alone. The studies show that visual presentation, message style, content

characteristics, and different types of engagement can play a role in how consumers respond to social media content. At the same time, the challenging studies show that engagement does not always lead to conversion because there are other factors that play a role in influencing a consumer. These factors can be purchase intention, brand attitude, information quality, and activity on other platforms. As a result of these challenges, AO3 should compare Facebook content and messaging strategies using both engagement metrics and OpenTable reservation activity instead of relying on one type of measure on its own.

Adequateness of Gathered Information

The information gathered for AO3 is adequate because the sources cover the key ideas that are important to the objective. These key ideas include Facebook content, messaging style, engagement, restaurant behavior, and booking or purchase intent. Some of the studies used are very relevant because they focus directly on Facebook or restaurant settings, like the work of Manningham, Asselin, and Bourguignon and Halloran and Lutz ([Manningham, Asselin, and Bourguignon 2024b](#)); ([Halloran and Lutz 2021b](#)). The research also used different methods, including content analysis, surveys, experiments, eye tracking, and regression analysis, which provides different types of evidence that can support AO3. There were also some studies that did not hold the same level of quality because the studies performed focused on hotels, Instagram, or other social media platforms instead of Facebook to restaurant activity. They also differed on what was being measured, such as purchase intention, visit intention, or booking intention rather than actual customer behavior. Overall, the research is strong enough to support AO3, but it does not fully answer whether Facebook engagement is connected to actual OpenTable activity, which shows why there is additional analysis that is needed.

Need for Additional Analysis

Additional analysis is still needed because the research gathered does not completely show how Facebook

affects customer response at RKR. The studies show that social media content and messaging can affect engagement, attention, and purchase or booking intent, but many of them don't measure actual reservations or visits. For my part of the project, I will focus on RKR's Facebook data by comparing different content types, messaging styles, and engagement metrics with available OpenTable Activity. Trackable links or QR codes can also help show if Facebook activity is connected to reservation behavior at RKR. Overall, this analysis will show what role Facebook may have within RKR's larger digital marketing strategy.

Assessment and Affirmation of Analytics Objectives

Based on the reviewed research, AO3 should remain the same because it directly supports RKR's larger goal of building sustainable demand. AO3 focuses on Facebook and whether specific content and messaging strategies are connected to stronger engagement and OpenTable reservation activity.

To complete AO3, I plan to organize RKR's Facebook posts into different content and messaging categories and compare how they perform using metrics such as likes, comments, shares, link clicks. I can then compare these results with available OpenTable activity to see if specific Facebook Strategies are also connected to stronger reservation related behavior. These methods allow me to look at both Facebook engagement and customer behavior instead of only using social media metrics. This approach also fits with the larger project because it uses both social media performance and reservation data.

2.1.4 AO4. Website Content, Navigation, and Usability (Georgia Monjarrez)

Existence of Well-established Models

Website usability is relevant to RKR because prospective guests need practical information before deciding whether and when to visit. This objective examines three specific activities: finding the menu, identifying operating hours, and reaching the reservation process. The literature provides complementary perspectives

on technology acceptance, information quality, and usability measurement. Together, these perspectives explain why the study should assess both what visitors can accomplish and how they evaluate the experience. The following applications to RKR are proposed research choices rather than findings about its current website.

Perceived ease of use and usefulness. Davis's technology acceptance research distinguishes perceived usefulness from perceived ease of use and develops measures of both constructs. The study reports relationships between these perceptions and system use, with usefulness showing a stronger relationship than ease of use (Davis 1989). This distinction matters because a website can feel simple to navigate while failing to provide information that helps a visitor make a decision. Conversely, useful information may require substantial effort to find. The original research concerns information technology acceptance rather than restaurant reservations, so it provides a theoretical starting point rather than direct evidence about RKR guests.

For this project, perceived ease will be assessed immediately after each task by asking how easy or difficult it was to complete. A separate overall item will ask whether the website provided the information needed to plan a visit. These questions address different judgments: effort and usefulness. For example, locating a menu quickly does not establish whether its contents meet a visitor's information needs. The proposed brief items adapt the concepts to RKR; they will not be described as the complete validated Davis scales. Pilot feedback should establish whether participants understand the wording and response options.

Information quality and hospitality decisions. Jeong and Lambert evaluated an information-quality framework using eight hypothetical lodging websites and 240 conference attendees. Their findings identified perceived usefulness and attitudes as significant indicators of customers' purchase-related behavior within that study (Jeong and Lambert 2001). This hospitality application supports examining information as part of a visitor's decision process, rather than treating a website only as a promotional display. How-

ever, simulated hotel websites and conference attendees differ from a student-operated restaurant and its prospective guests. The study therefore motivates local investigation without establishing that the same relationships or effect sizes apply to RKR.

For RKR, the proposed assessment will distinguish finding a page from finding the correct information. A menu task will require participants to identify the relevant menu, and an hours task will require them to state the posted hours for a specified service period. Before testing, the research team will verify the correct answers against the current website and record the testing date. This procedure should help distinguish a navigation problem from unclear or inconsistent content. Participant comments can identify information they expected but could not find; the team should verify those observations before recommending a content change.

Information systems success. DeLone and McLean's updated information systems success model provides a broader framework for evaluating systems and discusses its application to electronic commerce (DeLone and McLean 2003). Its relevance here is that website success requires an explicit definition of the outcomes being evaluated. AO4 will focus on a limited part of the customer journey: successful access to dining information and the reservation interface. It will not claim to test the entire information systems success model.

This boundary is important for interpreting the proposed results. A participant who reaches the reservation interface has completed the navigation task, even if no suitable table is available. A participant who reports that the website is helpful has expressed a perception, not made a booking. Reservation completion, revenue, and repeat patronage require additional evidence beyond this usability exercise. Consequently, the project objective is to recommend improvements that support access; any later claim that those improvements increase reservations would require implementation and follow-up measurement.

Usability as performance and perception. Hornbæk's review of 180 usability studies emphasizes the

need to distinguish and compare objective performance measures with subjective evaluations. It also highlights problems with treating usability too narrowly or using measures without sufficient justification (Hornbæk 2006). This supports collecting several complementary measures for AO4: task success, elapsed time, and perceived ease. These measures should be reported separately because a visitor can succeed slowly, fail quickly, or complete a task while finding the process frustrating.

The proposed study will calculate unassisted completion as the number of participants who finish a task correctly without help divided by the number who attempt it. Median completion time in seconds will summarize successful, unassisted attempts; assisted attempts, unsuccessful attempts, and timeouts will be reported separately. Immediately after every attempted task, participants will rate ease from 1 (very difficult) to 5 (very easy). Navigation observations and a short open-ended question will help explain the quantitative results. These operational definitions make the objective measurable while preserving the distinction between observed behavior and self-reported experience.

Adequateness of Gathered Information

The selected scholarly sources provide a foundation for choosing constructs and measures, but they cannot establish how well the current RKR website serves its users. They include foundational information-systems research, a usability review, and hospitality website research; their age and differing contexts limit direct transfer to contemporary mobile browsing and RKR's audience. Existing evidence is therefore adequate for developing the study but insufficient for answering AO4. Primary observations and post-task survey responses are needed from prospective guests, including campus and community participants where feasible. Prior familiarity with RKR and the website should be recorded because experienced users may navigate differently from first-time visitors.

Need for Additional Analysis

Participants will attempt the three tasks from the same starting page, using neutral instructions that do not reveal where to click. The team will pilot the procedure, set a consistent time limit before main data collection, vary task order across participants, and record device type and prior website familiarity. Timing will stop at success, a request for help, abandonment, or the time limit. The reservation task will end when the participant reaches the interface for beginning a reservation; no purchase or booking is required. For each task, analysis will report the success numerator and denominator, completion percentage, median successful completion time, and distribution of ease ratings, alongside recurring navigation problems. A proposed benchmark of 80% unassisted completion will be applied separately to each task; this is a project planning target, not a threshold established by the cited studies. Missing responses and technical failures will be documented separately from navigation failures.

Assessment and Affirmation of Analytics Objectives

AO4 remains appropriate because it connects PO4 to three observable tasks and defined measures. The reviewed frameworks support distinguishing useful information, perceived ease, and observed performance. They do not establish which changes RKR needs; primary testing will supply that evidence. Two descriptive research questions guide the assessment: How successfully can prospective guests locate the menu, identify operating hours, and reach the reservation interface, and does each task meet the proposed 80% unassisted completion target? Which content or navigation difficulties accompany unsuccessful attempts, longer successful completion times, or lower ease ratings?

Recommendations will link each observed problem to a specific proposed change and a corresponding measure for follow-up testing. For example, difficulty locating hours may justify clearer placement or labeling, while trouble reaching the reservation widget may justify a more visible entry point. These are possible recommendations to evaluate rather than conclusions about the current website. A later test using

the same task definitions would be needed to assess improvement after implementation.

The objective is primarily descriptive, so this review does not require separate causal hypotheses. Results from a convenience sample will describe the tested participants, with device type and prior familiarity considered during interpretation. The three tasks completed by one participant will not be treated as three independent participants. Successful navigation, perceived usefulness, completed reservations, and revenue remain distinct outcomes. This scope allows AO4 to support the group's reservation goal while keeping its conclusions proportional to the data collected.

2.1.5 AO5. Instagram Content and Engagement Analysis (Spencer Conway)

Analytics Objective: This analytics objective will determine which Instagram content format produces the highest engagement rate for Restaurant at Kellogg Ranch. First, past posts will be grouped by format to establish a performance baseline. RKR will then test new content in selected formats and compare its performance with the baseline. Engagement rate will be calculated consistently using likes, comments, shares, and saves divided by reach, where Instagram Insights data are available. The findings will guide recommendations about which formats RKR should prioritize in future Instagram posts.

Existence of Well-established Models

There are a few theories and research approaches that can help support AO5. The main ideas that connect to this objective are customer engagement, the Stimulus-Organism-Response model, inspiration, and visual presentation on social media. These ideas help explain why different Instagram posts or formats may get different reactions from customers.

Lee and Kwak (2026) studied restaurant social media and found that the way food is shown can affect customer inspiration and engagement.(Lee and Kwak 2026) Outcome-focused food images performed better

than process-focused images in their study. This is useful for AO5 because it shows that the way restaurant content is presented can make a difference in how people respond to it.

Bushara et al. (2023) used the Stimulus-Organism-Response model to study restaurant social media. (Bushara et al. 2023) Their study found that things like entertainment, interaction, customization, and trendiness can affect perceived value and customer behavior. This connects to AO5 because RKR's Instagram posts can be viewed as the stimulus and likes, comments, shares, and saves can be viewed as customer responses. One limitation is that Instagram Insights does not tell us exactly what a customer was thinking or feeling before engaging with a post.

Philp, Jacobson, and Pancer (2022) also looked directly at restaurant food marketing on Instagram. (Philp, Jacobson, and Pancer 2022b) They found that the visual characteristics of food images were connected to likes and comments. This supports AO5 because it shows that actual Instagram post data can be used to compare different types of content and see which characteristics are connected to stronger engagement.

Kulikovskaja et al. (2023) found that different types of social media content can lead to different types of customer engagement. (Kulikovskaja et al. 2023) This is important because RKR should not treat every Instagram post the same way. A Reel, carousel, or single-image post may give customers a different experience and may also create different levels of engagement.

Chan, Chen, and Leung (2023) also found that visual presentation can affect how customers respond to restaurant social media. (chan2023?) Their research showed that factors such as visual variety and movement can make a difference. This relates to AO5 because Instagram formats have different visual characteristics. For example, a Reel uses movement and video, while a regular post is more static.

Overall, the research supports the idea that the way restaurant social media content is presented can affect customer engagement. However, the existing research does not tell us exactly which Instagram format will work best for RKR. This is why analyzing RKR's own Instagram data is still needed.

Adequateness of Gathered Information

The information gathered so far is enough to support the direction of AO5. The research shows that restaurant social media content, visual presentation, and content type can affect engagement. Some of the studies also focus directly on Instagram or restaurant social media, which makes them useful for this project. However, the research cannot tell us which format works best specifically for RKR. Different restaurants have different audiences and different types of content, so RKR's own Instagram data still needs to be analyzed.

Need for Additional Analysis

Additional analysis is needed to compare RKR's actual Instagram performance. I plan to group past posts into formats such as single-image posts, carousels, and Reels or videos. I will then collect available engagement metrics such as likes, comments, shares, saves, and reach. Engagement rate will be calculated using the same method across posts so the formats can be compared more fairly.

After creating a baseline from past posts, RKR can also test new content in selected formats and compare the results with the historical performance. This should help show whether certain Instagram formats consistently perform better for RKR.

Assessment and Affirmation of Analytics Objectives

Based on the research, AO5 should remain the same. The research supports the idea that different social media formats and visual characteristics can affect customer engagement, but there is still a need to test this using RKR's own Instagram data.

To complete AO5, I plan to compare RKR's Instagram formats using engagement rate. Posts will be grouped by format and compared using likes, comments, shares, saves, and reach where the data is avail-

able. The research models will help explain why some formats may perform differently, while the actual Instagram data will be used to determine which formats work best for RKR.

The final goal is to give RKR a more data-driven Instagram strategy and recommend which formats they should prioritize moving forward. The results will focus on Instagram engagement and will not assume that higher engagement automatically means more restaurant reservations or sales.

References

- Arora, Raj and Joe Singer (2006), “[Customer satisfaction and value as drivers of business success for fine dining restaurants](#),” *Services Marketing Quarterly*, 28 (1), 89–102.
- Bashirzadeh, Yashar, Robert Mai, and Corinne Faure (2022), “[How rich is too rich? Visual design elements in digital marketing communications](#),” *International Journal of Research in Marketing*, 39 (1), 58–76.
- Bicudo De Castro, Vicente (2025), “[Quantifying the burden of organisational bulk emails in a business school](#),” *Higher Education Research & Development*, 44 (8), 1934–48.
- Bushara, Meimona Abdelrhim, Ahmed Hassan Abdou, Thowayeb H. Hassan, Abu Elnasr E. Sobaih, Abdullah Saleh Mohammed Albohnayh, Waleed Ghazi Alshammari, Mohammed Aldoreeb, Ahmed Anwar Elsaed, and Mohamed Ahmed Elsaied (2023), “[Power of Social Media Marketing: How Perceived Value Mediates the Impact on Restaurant Followers’ Purchase Intention, Willingness to Pay a Premium Price, and E-WoM?](#)” *Sustainability*, 15 (6), 5331.
- Davis, Fred D. (1989), “[Perceived usefulness, perceived ease of use, and user acceptance of information technology](#),” *MIS Quarterly*, 13 (3), 319–40.
- Dedeke, Adenekan Nick (2016), “[Travel web-site design: Information task-fit, service quality and purchase intention](#),” *Tourism Management*, 54, 541–54.

- DeLone, William H. and Ephraim R. McLean (2003), “[The DeLone and McLean model of information systems success: A ten-year update](#),” *Journal of Management Information Systems*, 19 (4), 9–30.
- Go, Sydney (2024), “[What Is Email Marketing? The Definitive Guide for Beginners](#).”
- Halloran, Timothy J. and Richard J. Lutz (2021), “[Let’s Give Them Something to Talk About: Which Social Media Engagements Predict Purchase Frequency?](#)” *Journal of Interactive Marketing*, 56 (1), 83–95.
- Han, Heesup and Kisang Ryu (2009), “[The roles of the physical environment, price perception, and customer satisfaction in determining customer loyalty in the restaurant industry](#),” *Journal of Hospitality & Tourism Research*, 33 (4), 487–510.
- Hanaysha, Jalal Rajeh (2022), “[An examination of social media advertising features, brand engagement and purchase intention in the fast food industry](#),” *British Food Journal*, 124 (11), 4143–60.
- Hartemo, Mari, Reima Suomi, and Ulla Hakala (2016), “[Towards improved performance: A model for testing email newsletter design](#),” *Journal of Electronic Commerce in Organizations*, 14 (3), 1–16.
- Heung, Vincent C. S. and Tianming Gu (2012), “[Influence of restaurant atmospherics on patron satisfaction and behavioral intentions](#),” *International Journal of Hospitality Management*, 31 (4), 1167–77.
- Hornbæk, Kasper (2006), “[Current practice in measuring usability: Challenges to usability studies and research](#),” *International Journal of Human-Computer Studies*, 64 (2), 79–102.
- Hornbæk, Kasper and Effie Lai-Chong Law (2007), “[Meta-analysis of correlations among usability measures](#),” in *Proceedings of the SIGCHI conference on human factors in computing systems*, Association for Computing Machinery, 617–26.
- Jeong, Miyoung and Carolyn U. Lambert (2001), “[Adaptation of an information quality framework to measure customers’ behavioral intentions to use lodging web sites](#),” *International Journal of Hospitality Management*, 20 (2), 129–46.
- Jones, Alexandra (2023), “[The complete restaurant email marketing guide](#).”

- Kim, Hak-Seon, Kwang-Woo Lee, Dong-Soo Lee, Hyun-Woo Joung, and Jingxue Jessica Yuan (2012), “[Assessing the quality of a restaurant’s website using DINEWEBQUAL](#),” *Journal of Quality Assurance in Hospitality & Tourism*, 13 (3), 235–45.
- Kong, Ruoyan, Haiyi Zhu, and Joseph A. Konstan (2021), “[Learning to Ignore: A Case Study of Organization-Wide Bulk Email Effectiveness](#),” *Proceedings of the ACM on Human-Computer Interaction*, 5 (CSCW1), 1–23.
- Kulikovskaja, Viktorija, Marco Hubert, Klaus G. Grunert, and Hong Zhao (2023), “[Driving marketing outcomes through social media-based customer engagement](#),” *Journal of Retailing and Consumer Services*, 74, 103445.
- Kumar, Ashish (2021), “[An empirical examination of the effects of design elements of email newsletters on consumers’ email responses and their purchase](#),” *Journal of Retailing and Consumer Services*, 58, 102349.
- Kumar, Ashish and Jari Salo (2018), “[Effects of link placements in email newsletters on their click-through rate](#),” *Journal of Marketing Communications*, 24 (5), 535–48.
- Lee, Seonjeong (Ally) and Soo Yeon Kwak (2026), “[Enhancing customer inspiration and engagement experiences on restaurant social media](#),” *International Journal of Hospitality Management*, 138, 104713.
- Manningham, Daphnée, Hugo Asselin, and Benoit Bourguignon (2024), “[Be Direct! Restaurant Social Media Posts to Drive Customer Engagement in Times of Crisis and Beyond](#),” *Tourism and Hospitality*, 5 (2), 304–13.
- Namkung, Young and SooCheong Jang (2007), “[Does food quality really matter in restaurants? Its impact on customer satisfaction and behavioral intentions](#),” *Journal of Hospitality & Tourism Research*, 31 (3), 387–409.
- Philp, Matthew, Jenna Jacobson, and Ethan Pancer (2022b), “[Predicting social media engagement with computer vision: An examination of food marketing on Instagram](#),” *Journal of Business Research*,

149, 736–47.

Philp, Matthew, Jenna Jacobson, and Ethan Pancer (2022a), “[Predicting social media engagement with computer vision: An examination of food marketing on Instagram](#),” *Journal of Business Research*, 149, 736–47.

Ryu, Kisang, Hye-Rin Lee, and Woo Gon Kim (2012), “[The influence of the quality of the physical environment, food, and service on restaurant image, customer perceived value, customer satisfaction, and behavioral intentions](#),” *International Journal of Contemporary Hospitality Management*, 24 (2), 200–223.

Skourou, Evangelia and Dimitris Spiliotopoulos (2024), “[Assessment of purchasing influence of email campaigns using eye tracking](#),” *Multimodal Technologies and Interaction*, 8 (10), 87.

Tuch, Alexandre N., Sandra P. Roth, Kasper Hornbæk, Klaus Opwis, and Javier A. Bargas-Avila (2012), “[Is beautiful really usable? Toward understanding the relation between usability, aesthetics, and affect in HCI](#),” *Computers in Human Behavior*, 28 (5), 1596–1607.

Veloso, Mónica and Monica Gomez-Suarez (2023), “[The influential role of hotel-generated content on social media](#),” *Journal of Hospitality and Tourism Technology*, 14 (2), 245–57.

Yost, Elizabeth, Tingting Zhang, and Ruoxi Qi (2021), “[The power of engagement: Understanding active social media engagement and the impact on sales in the hospitality industry](#),” *Journal of Hospitality and Tourism Management*, 46, 83–95.

Zunac, Ana Globocnik, Lana Miletic, and Goran Kozina (2022), “[Self-sufficiency of e-mail as a marketing tool](#),” Varazdin, Croatia: Varazdin Development; Entrepreneurship Agency (VADEA), 147–53.